Sequences of random variables

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Sequences of random variables

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The weak law of large numbers

Theorem (WLLN)

Let $X_1, X_2, \ldots, X_k, \ldots$ be a sequence of independent and identically distributed random variables, each having a finite expectation m and finite variance.

Set

$$\overline{X}_n = \frac{X_1 + X_2 + \cdots + X_n}{n}, \quad n \geqslant 1.$$

Then, for all $\epsilon > 0$,

$$\lim_{n \to \infty} \mathbb{P}\left(\left|\overline{X}_n - m\right| \geqslant \epsilon\right) = 0$$

The weak law of large numbers

Proof: We know that

$$\mathbb{E}\left(\overline{X}_n\right) = m, \quad \mathrm{Var}\left(\overline{X}_n\right) = \frac{\mathrm{Var}(X_1)}{n}$$

Applying Chebyshev's inequality,

$$\mathbb{P}\left(\left|\overline{X}_n - m\right| \geqslant \epsilon\right) \leqslant \frac{\mathrm{Var}(X_1)}{n\,\epsilon^2} \to 0 \quad \text{as } n \to \infty$$

- The sequence $\left\{\overline{X}_n \colon n \geqslant 1\right\}$ converges in a specified sense to the limit value $m = \mathbb{E}(X_1)$.

Convergence in probability

Definition

The sequence of random variables $X_1, X_2, \ldots, X_n, \ldots$ converges in probability to the random variable X if for all $\epsilon > 0$,

$$\mathbb{P}\left(|X_n - X| \geqslant \epsilon\right) \to 0 \quad \text{as } n \to \infty,$$

or, equivalently,

$$\mathbb{P}\left(|X_n - X| < \epsilon\right) \to 1 \quad \text{as } n \to \infty$$

Notation: $X_n \xrightarrow{\mathbb{P}} X$.

- For the sequence of sample means $\left\{\overline{X}_n \colon n \geqslant 1\right\}$ we have $\overline{X}_n \xrightarrow{\mathbb{P}} m$.

Example

Let $X_1, X_2, \ldots, X_n, \ldots$ be a sequence of discrete random variables such that

$$\mathbb{P}(X_n = 0) = 1 - \frac{1}{n}$$

$$\mathbb{P}(X_n = -1) = \mathbb{P}(X_n = 1) = \frac{1}{2n}, \quad n \geqslant 1$$

Let us show that

$$X_n \xrightarrow{\mathbb{P}} 0$$

Example

We have to prove that, for all $\epsilon > 0$, $\mathbb{P}(|X_n - 0| \geqslant \epsilon) \to 0$ as $n \to \infty$.

Equivalently, let us show that

$$\mathbb{P}(|X_n| < \epsilon) \longrightarrow 1 \quad \text{as } n \to \infty$$

- If $\epsilon > 1$, then $\mathbb{P}(|X_n| < \epsilon) = 1$ for all $n \geqslant 1$.
- Otherwise, if $\epsilon \leqslant 1$, then

$$\mathbb{P}(|X_n| < \epsilon) = \mathbb{P}(X_n = 0) = 1 - \frac{1}{n} \longrightarrow 1$$

Probability as the limit of the relative frequency

A significant implication of the WLLN is as follows:

- Let A be an event with probability $\mathbb{P}(A)$.
- In each of n independent trials, we observe whether or not A occurs.

  More precisely, for $1 \leqslant k \leqslant n$, let X_k be the indicator of the event "A happens in the k-th repetition".

Hence, $\mathbb{E}(X_k) = \mathbb{P}(A)$ and

$$\overline{X}_n = \frac{X_1 + X_2 + \cdots + X_n}{n} = f_n(A)$$

is the relative frequency of the event A.

Probability as the limit of the relative frequency

Corollary

The sequence of relative frequencies $\{f_n(A) \colon n \geqslant 1\}$ *converges in probability to* $\mathbb{P}(A)$*:*

$$f_n(A) \xrightarrow{\mathbb{P}} \mathbb{P}(A)$$

That is to say, for all $\epsilon > 0$*,*

$$\mathbb{P}\left(|f_n(A) - \mathbb{P}(A)| < \epsilon\right) \to 1 \quad \text{as } n \to \infty$$

The standardized sample mean

Let $X_1, X_2, \ldots, X_k, \ldots$ be a sequence of independent and identically distributed random variables such that $\mathbb{E}(X_k) = m$ and $\mathrm{Var}(X_k) = \sigma^2$.

Let us consider the sequence of the standardized sample means:

$$Z_n = \frac{\overline{X}_n - \mathbb{E}(\overline{X}_n)}{\sqrt{\mathrm{Var}(\overline{X}_n)}} = \frac{\overline{X}_n - m}{\sigma/\sqrt{n}}$$

$$= \frac{1}{\sqrt{n}} \sum_{k=1}^{n} \frac{X_k - m}{\sigma} = \frac{(X_1 + \cdots + X_n) - n\,m}{\sqrt{n}\,\sigma}, \quad n \geqslant 1$$

Observe that $\mathbb{E}(Z_n) = 0$, $\mathrm{Var}(Z_n) = 1$, and $\overline{X}_n = (\sigma/\sqrt{n})\, Z_n + m$

The standardized sample mean

If $M_Z(\omega)$ is the common characteristic function of the variables $Z_k = (X_k - m)/\sigma$, then the characteristic function $M_n(\omega)$ of Z_n is given by

$$
\begin{aligned}
M_n(\omega) &= \mathbb{E}\left(e^{i\omega Z_n}\right) \\
&= \mathbb{E}\left(e^{i\omega \frac{1}{\sqrt{n}} \sum_{k=1}^{n} \frac{X_k - m}{\sigma}}\right) = \mathbb{E}\left(\prod_{k=1}^{n} e^{i \frac{\omega}{\sqrt{n}} \frac{X_k - m}{\sigma}}\right) \\
&= \prod_{k=1}^{n} \mathbb{E}\left(e^{i \frac{\omega}{\sqrt{n}} \frac{X_k - m}{\sigma}}\right) = \left(M_Z\left(\frac{\omega}{\sqrt{n}}\right)\right)^n
\end{aligned}
$$

The standardized sample mean

The first terms of the series expansion of $M_Z(u)$ are:

$$\begin{aligned} M_Z(u) &= 1 + i\,\mathbb{E}(Z)u + \frac{i^2\,\mathbb{E}(Z^2)}{2}u^2 + o(u^2) \\ &= 1 - \frac{1}{2}u^2 + o(u^2), \end{aligned}$$

where $o(u^2)$ denotes a residual term such that

$$\frac{o(u^2)}{u^2} \to 0 \quad \text{as } u \to 0$$

The standardized sample mean

Therefore,

$$
\begin{aligned}
M_n(\omega) &= \left(M_Z\left(\frac{\omega}{\sqrt{n}}\right)\right)^n \\
&= \left(1 - \frac{1}{2}\left(\frac{\omega}{\sqrt{n}}\right)^2 + o\left(\frac{\omega}{\sqrt{n}}\right)^2\right)^n \\
&= \left(1 - \frac{1}{2}\cdot\frac{\omega^2}{n} + o\left(\frac{1}{n}\right)\right)^n
\end{aligned}
$$

- We want to analyse the behaviour of $M_n(\omega)$ as n increases.

The standardized sample mean

Taking logarithms and using that $\ln(1+z) = z + o(|z|)$,

$$\begin{aligned}
\ln M_n(\omega) &= n \cdot \ln\left(1 - \frac{1}{2} \cdot \frac{\omega^2}{n} + o\left(\frac{1}{n}\right)\right) \\
&= n\left(-\frac{1}{2} \cdot \frac{\omega^2}{n} + o\left(\frac{1}{n}\right)\right) \\
&= -\frac{1}{2}\,\omega^2 + \frac{o(1/n)}{1/n} \longrightarrow -\frac{1}{2}\,\omega^2 \quad \text{as } n \to \infty
\end{aligned}$$

The standardized sample mean

We have proved that

$$M_n(\omega) \to e^{-\omega^2/2} \quad \text{as } n \to \infty,$$

where

- *the convergence is pointwise for all* $\omega \in \mathbb{R}$,
- *the limit function* $M(\omega) = e^{-\omega^2/2}$ *is the characteristic function of a standard normal random variable.*

This type of convergence regarding characteristic functions can also be expressed in terms of distribution functions. We require a result known as the continuity theorem.

The continuity theorem

Let $\{F_n(x)\colon n \geqslant 1\}$ be a sequence of probability distribution functions, and let $\{M_n(\omega)\colon n \geqslant 1\}$ be the sequence of the corresponding characteristic functions.

Theorem (Continuity Theorem)

(a) *If $M(\omega) = \lim_{n\to\infty} M_n(\omega)$ exists and is continuous at $\omega = 0$, then $M(\omega)$ is the characteristic function of some distribution function $F(x)$, and $F_n(x) \to F(x)$ at each point x where the limit function $F(x)$ is continuous.*

(b) *If for some distribution function $F(x)$, with characteristic function $M(\omega)$, we have that $F_n(x) \to F(x)$ at each point x where $F(x)$ is continuous, then $M_n(\omega) \to M(\omega)$ for all $\omega \in \mathbb{R}$.*

The central limit theorem

Theorem (CLT)

Let $X_1, X_2, \ldots, X_k, \ldots$ *be a sequence of independent and identically distributed random variables such that* $\mathbb{E}(X_k) = m$ *and* $\mathrm{Var}(X_k) = \sigma^2$*. Let*

$$Z_n = \frac{1}{\sqrt{n}} \sum_{k=1}^{n} \frac{X_k - m}{\sigma}, \quad n \geqslant 1$$

Then, for all $x \in \mathbb{R}$*,*

$$\lim_{n\to\infty} F_{Z_n}(x) = F_{\mathrm{N}(0,1)}(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2}\, dt$$

Convergence in distribution

Definition

(a) A sequence $F_1, F_2, \ldots, F_n, \ldots$ of distribution functions converges to the distribution function F, written $F_n \to F$, if $F_n(x) \to F(x)$ at each point x where the limit function F is continuous.

(b) We say that the sequence of random variables $X_1, X_2, \ldots, X_n, \ldots$ converges in distribution to the random variable X if $F_{X_n} \to F_X$.

Notation: $X_n \xrightarrow{d} X$.

Example. The Central Limit Theorem:

$$\frac{1}{\sqrt{n}} \sum_{k=1}^{n} \frac{X_k - m}{\sigma} \xrightarrow{d} \mathrm{N}(0,1) \quad \text{as } n \to \infty$$

The central limit theorem

(Remark)

The CLT is sometimes expressed by saying that $\overline{X}_n$ is asymptotically normally distributed with mean m and variance σ^2/n, in the following sense:

For all $x \in \mathbb{R}$,

$$\lim_{n\to\infty} \left| \mathbb{P}\left(\overline{X}_n \leqslant x\right) - \frac{1}{\sqrt{2\pi\frac{\sigma^2}{n}}} \int_{-\infty}^{x} e^{-\frac{(t-m)^2}{2\sigma^2/n}} \right| = 0$$

The second term is $\mathbb{P}(Y_n \leqslant x)$, being $Y_n \sim \mathrm{N}\left(m, \frac{\sigma^2}{n}\right)$.

WLLN versus CLT

Example:

Let $X_1, X_2, \ldots, X_n, \ldots$ be a sequence of independent random variables, each one uniformly distributed in $[0, 1]$.

- Since $\mathbb{E}(\overline{X}_n) = 1/2$, by the WLLN we know that, given $\alpha > 0$,

$$\mathbb{P}\left(\left|\overline{X}_n - \frac{1}{2}\right| > \frac{1}{\alpha}\right) \longrightarrow 0 \quad \text{as } n \to \infty$$

- The CLT offers more precise quantitative information. Given that $\mathrm{Var}(X_n) = 1/12$, it follows that

$$\frac{\overline{X}_n - \mathbb{E}(\overline{X}_n)}{\sqrt{\mathrm{Var}(\overline{X}_n)}} = \frac{\overline{X}_n - 1/2}{1/\sqrt{12n}} \xrightarrow{d} \mathrm{N}(0, 1)$$

WLLN versus CLT

- Therefore,

$$\mathbb{P}\left(\left|\overline{X}_n - \frac{1}{2}\right| > \frac{1}{\alpha}\right) = 1 - \mathbb{P}\left(-\frac{1}{\alpha} \leqslant \overline{X}_n - \frac{1}{2} \leqslant \frac{1}{\alpha}\right)$$

$$= 1 - \mathbb{P}\left(-\frac{\sqrt{12n}}{\alpha} \leqslant \frac{\overline{X}_n - 1/2}{1/\sqrt{12n}} \leqslant \frac{\sqrt{12n}}{\alpha}\right)$$

$$\approx 1 - \left(F_{\mathrm{N}(0,1)}\left(\frac{\sqrt{12n}}{\alpha}\right) - F_{\mathrm{N}(0,1)}\left(-\frac{\sqrt{12n}}{\alpha}\right)\right)$$

$$= 2\left(1 - F_{\mathrm{N}(0,1)}\left(\frac{\sqrt{12n}}{\alpha}\right)\right)$$

De Moivre-Laplace Theorem

If each X_k is a Bernoulli random variable with parameter p, then

$$S_n = X_1 + X_2 + \cdots + X_n$$

follows a $\mathrm{Bin}(n, p)$ distribution with expected value np and variance npq.

Hence, the CLT implies

Theorem (De Moivre-Laplace)

Let $S_n \sim \mathrm{Bin}(n, p)$. Then

$$\frac{S_n - np}{\sqrt{npq}} \xrightarrow{d} \mathrm{N}(0, 1) \quad \textit{as } n \to \infty$$

De Moivre-Laplace Theorem

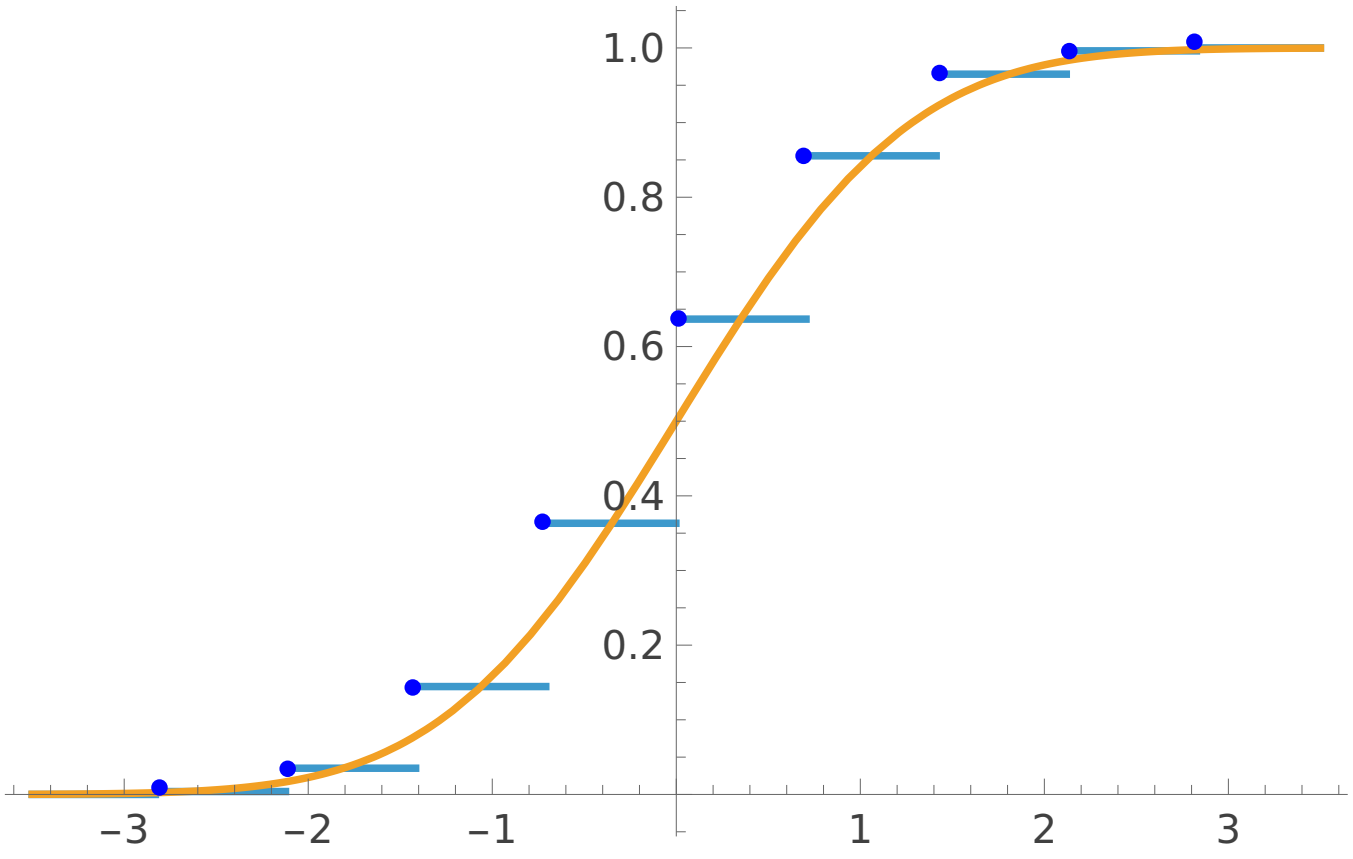

Figure: Normalized $\mathrm{Bin}(n, 1/2)$ for $n = 8$, $F_{\mathrm{N}(0,1)}$

A local version of the CLT

Theorem

Let $X_1, X_2, \ldots, X_k, \ldots$ be independent and identically distributed random variables with zero mean, unit variance, and common characteristic function $M_X(\omega)$. Let

$$U_n = \frac{1}{\sqrt{n}}\left(X_1 + X_2 + \cdots + X_n\right)$$

If $\int_{-\infty}^{\infty} |M_X(\omega)|^r \; d\omega < \infty$ for some integer $r \geqslant 1$, then the density function $f_n(u)$ of the variable U_n exists for all $n \geqslant r$, and, moreover,

$$f_n(u) \to \frac{1}{\sqrt{2\pi}}\, e^{-u^2/2} \quad \text{as } n \to \infty,$$

where the convergence is uniform on $\mathbb{R}$.

Example: sum of uniform random variables

Suppose that each X_k is uniformly distributed on the interval $\left[-\sqrt{3}, \sqrt{3}\right]$. (Hence, $\mathbb{E}(X_k) = 0$ and $\mathrm{Var}(X_k) = 1$.)

The common characteristic function is:

$$M_X(\omega) = \int_{-\infty}^{\infty} e^{i\omega x} f_X(x)\, dx = \frac{1}{2\sqrt{3}} \int_{-\sqrt{3}}^{\sqrt{3}} e^{i\omega x}\, dx$$

$$= \frac{1}{\sqrt{3}\,\omega} \frac{e^{i\sqrt{3}\,\omega} - e^{-i\sqrt{3}\,\omega}}{2i}$$

$$= \frac{\sin\left(\sqrt{3}\,\omega\right)}{\sqrt{3}\,\omega}$$

Example: sum of uniform random variables

It can be shown that

$$\int_{-\infty}^{\infty} |M_X(\omega)|^2 \, d\omega = \int_{-\infty}^{\infty} \left(\frac{\sin\left(\sqrt{3}\,\omega\right)}{\sqrt{3}\,\omega} \right)^2 d\omega = \frac{\pi}{\sqrt{3}} < \infty,$$

ando so the sufficient condition of the theorem holds for $r = 2$.

Hence the density f_n of the variable

$$U_n = \frac{X_1 + X_2 + \cdots + X_n}{\sqrt{n}}$$

exists for all $n \geqslant 1$ and

$$f_n(u) \to \frac{1}{\sqrt{2\pi}} \, e^{-u^2/2}$$

Example: sum of uniform random variables

For instance, let $U_3 = (X_1 + X_2 + X_3)/\sqrt{3}$.

The density function of the variable U_3 can be expressed as

$$f_3(u) = \sqrt{3}\, f(\sqrt{3}\, u),$$

where $f(s) = f_X(s) * f_X(s) * f_X(s)$.

The calculation of $f_3(u)$ gives:

$$f_3(u) = \begin{cases} 0, & u < -3 \\ (u+3)^2/16, & -3 \leqslant u < -1 \\ (3-u^2)/8, & -1 \leqslant u < 1 \\ (3-u)^2/16, & 1 \leqslant u < 3 \\ 0, & u \geqslant 3 \end{cases}$$

Example: sum of uniform random variables

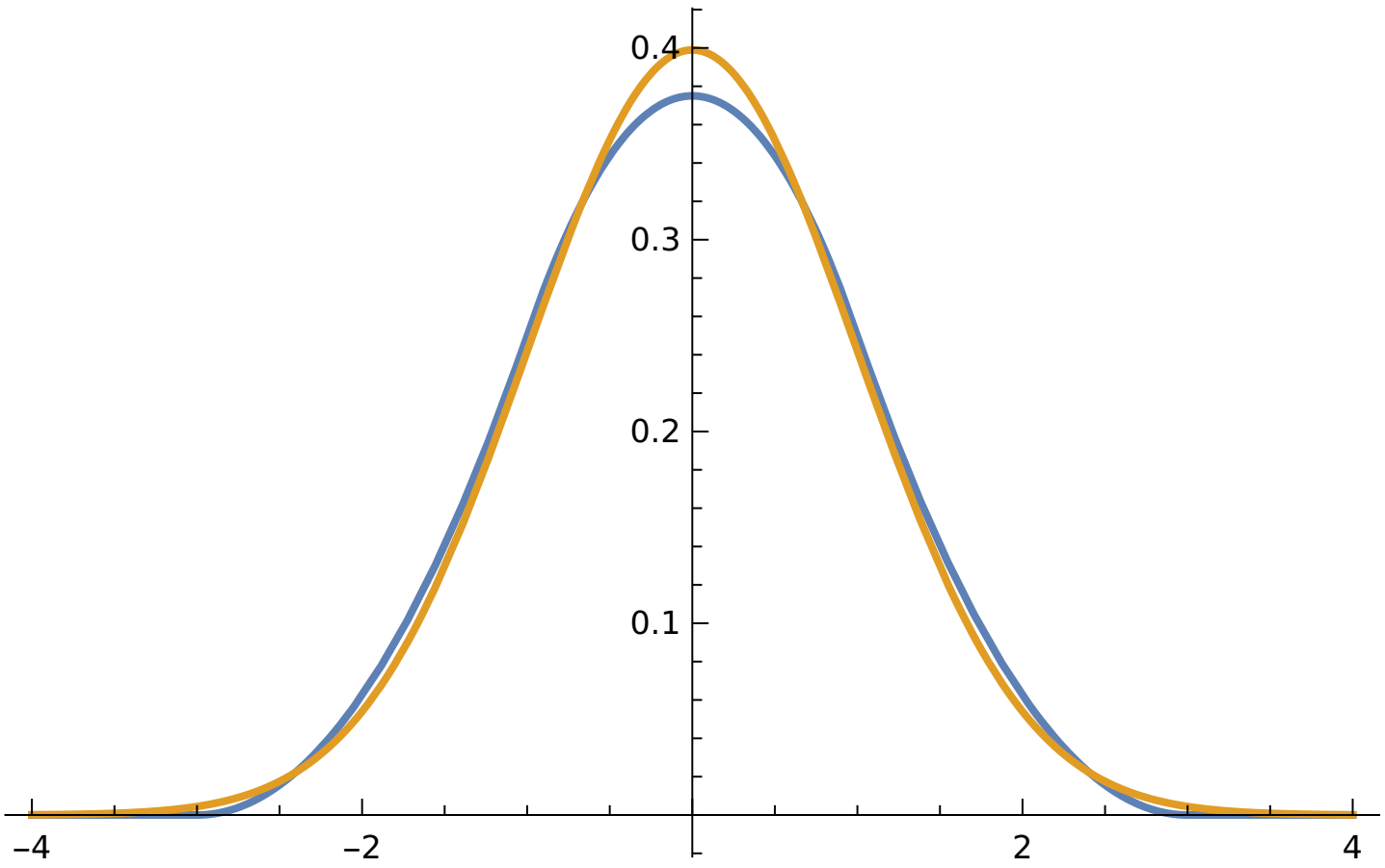

Figure: $f_3(u)$, $\frac{1}{\sqrt{2\pi}} e^{-u^2/2}$

Poisson's theorem

The following result shows another application of the continuity theorem.

Theorem (Poisson)

If $X_n \sim \text{Bin}\,(n, \lambda/n)$, *then* $X_n \xrightarrow{d} \text{Po}(\lambda)$.

Proof:

$$M_{X_n}(\omega) = \left(1 - \frac{\lambda}{n} + \frac{\lambda}{n}\, e^{i\omega}\right)^n$$

$$= \left(1 + \frac{\lambda(e^{i\omega} - 1)}{n}\right)^n \to e^{\lambda(e^{i\omega}-1)} \quad \text{as } n \to \infty$$

Convergence in distribution: a result

Theorem

Let $X_1, X_2, \ldots X_n \ldots$ and X be random variables taking nonnegative integer values. Then $X_n \xrightarrow{d} X$ if and only if

$$\lim_{n\to\infty} \mathbb{P}(X_n = k) = \mathbb{P}(X = k) \quad \textit{for all } k \geqslant 0$$

For example, by Poisson's theorem and this result, we have that for all $k \geqslant 0$,

$$\binom{n}{k}\left(\frac{\lambda}{n}\right)^k\left(1-\frac{\lambda}{n}\right)^{n-k} \longrightarrow e^{-\lambda}\,\frac{\lambda^k}{k!} \quad \text{as } n \to \infty$$

Convergence in mean square

Given a probability space, we can associate to it an abstract vector space $\mathcal{H}$:

The elements of $\mathcal{H}$ are random variables X satisfying $\mathbb{E}(X^2) < \infty$, indicating a finite second-order moment. The internal operation in $\mathcal{H}$ is the addition of random variables, while scalar multiplication by a real number constitutes the external operation.

- $\mathcal{H}$ can be equipped with an inner product: $\langle X, Y \rangle \equiv \mathbb{E}(XY)$.
- The induced norm is $\|X\| = \sqrt{\langle X, X \rangle} = \sqrt{\mathbb{E}(X^2)}$
- Moreover, we can define a distance between points in the set $\mathcal{H}$ as follows:

$$d(X, Y) = \|X - Y\| = \sqrt{\mathbb{E}\left((X - Y)^2\right)}$$

Convergence in mean square

Definition

The sequence $X_1, X_2, \ldots, X_n, \ldots$ converges in mean square to the random variable X if $d(X_n, X) \to 0$ as $n \to \infty$.

Equivalently, $\mathbb{E}\left((X_n - X)^2\right) \to 0$ as $n \to \infty$.

Notation: $X_n \xrightarrow{2} X$.

Example: The sequence of sample means $\overline{X}_n$ converges in mean square to the expected value m, because

$$\mathbb{E}\left(\left(\overline{X}_n - m\right)^2\right) = \mathrm{Var}\left(\overline{X}_n\right) = \frac{\mathrm{Var}(X_1)}{n} \to 0$$

Convergence in r-mean

More generally,

Definition

The sequence of random variables $X_1, X_2, \ldots, X_n, \ldots$ converges in r-mean to the random variable X if

$$\mathbb{E}\left(|X_n - X|^r\right) \to 0 \quad \text{as } n \to \infty$$

Notation: $X_n \xrightarrow{r} X$.

Example

Consider the sequence $X_1, X_2, \ldots, X_n, \ldots$, such that

$$\begin{cases} \mathbb{P}(X_n = 0) = 1 - \dfrac{1}{n} \\ \mathbb{P}(X_n = -1) = \mathbb{P}(X_n = 1) = \dfrac{1}{2n}, \quad n \geqslant 1 \end{cases}$$

Let us prove that $X_n \xrightarrow{r} 0$ for any $r > 0$.

Proof: We have to show that $\mathbb{E}\left(|X_n - 0|^r\right) \to 0$ as $n \to \infty$.

Indeed,

$$\begin{aligned} \mathbb{E}(|X_n|^r) &= 0 \cdot \mathbb{P}(X_n = 0) + 1 \cdot (\mathbb{P}(X_n = -1) + \mathbb{P}(X_n = 1)) \\ &= \frac{1}{2n} + \frac{1}{2n} \longrightarrow 0 \end{aligned}$$

Almost surely convergence

- Consider a sequence $X_1, X_2, \ldots, X_n, \ldots$ of random variables. For each outcome ω of the random experiment, the set $\{X_n(\omega) \colon n \geqslant 1\}$ constitutes a numerical sequence of real numbers.
- If X is a random variable and $X(\omega)$ is the numerical value taken by X for the outcome ω, we can consider the event:

$$\{\omega \in \Omega \colon X_n(\omega) \to X(\omega) \ \text{ as } n \to \infty\}$$

Almost surely convergence

Definition

The sequence $X_1, X_2, \ldots, X_n, \ldots$ converges almost surely (or with probability 1) to the random variable X if

$$\mathbb{P}\left(\{\omega \in \Omega \colon X_n(\omega) \to X(\omega) \ \text{ as } n \to \infty\}\right) = 1$$

Notation: $X_n \xrightarrow{a.s.} X$.

Notice that $X_n \xrightarrow{a.s.} X$ if and only if the set

$$N = \{\omega \in \Omega \colon X_n(\omega) \not\to X(\omega) \ \text{ as } n \to \infty\}$$

is a null event, that is to say, $\mathbb{P}(N) = 0$.

Strong law of large numbers

Theorem (SLLN)

Let $X_1, X_2, \ldots, X_k, \ldots$ be a sequence of independent and identically distributed random variables and set

$$\overline{X}_n = \frac{X_1 + X_2 + \cdots + X_n}{n}, \quad n \geqslant 1$$

Then

$$\overline{X}_n \xrightarrow{a.s.} m \quad \text{as } n \to \infty$$

for some constant m, if and only if $\mathbb{E}(|X_1|) < \infty$. In this case, $E(X_1) = m$.

Uniqueness

Theorem

Let $X_1, X_2, \ldots, X_n, \ldots$ be a sequence of random variables. If the sequence converges

- *almost surely,*
- *in probability,*
- *in r-mean,*
- *in distribution,*

then the limiting random variable (distribution) is unique

Relations between the convergence concepts

Theorem

The following implications hold:

- $\left(X_n \xrightarrow{a.s.} X\right) \Longrightarrow \left(X_n \xrightarrow{\mathbb{P}} X\right)$.
- $\left(X_n \xrightarrow{r} X\right) \Longrightarrow \left(X_n \xrightarrow{\mathbb{P}} X\right)$ *for any* $r \geqslant 1$.
- $\left(X_n \xrightarrow{\mathbb{P}} X\right) \Longrightarrow \left(X_n \xrightarrow{d} X\right)$.
- *If* $r > s \geqslant 1$, *then* $\left(X_n \xrightarrow{r} X\right) \Longrightarrow \left(X_n \xrightarrow{s} X\right)$.

All implications are strict.

Relations between the convergence concepts

With additional hypothesis some converse implications also hold.

Theorem

- *If c is a constant, then* $\left(X_n \xrightarrow{d} c\right) \Longrightarrow \left(X_n \xrightarrow{\mathbb{P}} c\right)$.
- *If* $X_n \xrightarrow{\mathbb{P}} X$ *and there exists a constant C such that* $\mathbb{P}(|X_n| \leqslant C) = 1$ *for all n, then* $X_n \xrightarrow{r} X$ *for all* $r \geqslant 1$.
- *If* $P_n(\epsilon) = \mathbb{P}\left(|X_n - X| > \epsilon\right)$ *satisfies* $\sum_n P_n(\epsilon) < \infty$ *for all* $\epsilon > 0$, *then* $X_n \xrightarrow{a.s.} X$.

Relations between the convergence concepts

For instance, let us consider the proof of the implication

$$\left(X_n \xrightarrow{d} c\right) \Longrightarrow \left(X_n \xrightarrow{\mathbb{P}} c\right)$$

Hence, assume that $X_n \xrightarrow{d} X$, where $X = c$ is a "constant" random variable with distribution function

$$F_X(x) = \begin{cases} 0, & x < c \\ 1, & x \geqslant c \end{cases}$$

Relations between the convergence concepts

If $\epsilon > 0$ is a fixed number, we have that

$$\begin{aligned} P(|X_n - c| > \epsilon) &= 1 - \mathbb{P}(c - \epsilon \leqslant X_n \leqslant c + \epsilon) \\ &= 1 - (F_{X_n}(c+\epsilon) - F_{X_n}(c-\epsilon) + \mathbb{P}(X_n = c - \epsilon)) \\ &\leqslant 1 - F_{X_n}(c+\epsilon) + F_{X_n}(c-\epsilon) \longrightarrow 0, \end{aligned}$$

because

$$F_{X_n}(c+\epsilon) \to F_X(c+\epsilon) = 1, \quad F_{X_n}(c-\epsilon) \to F_X(c-\epsilon) = 0$$

Therefore,

$$X_n \xrightarrow{\mathbb{P}} c$$

Example

Consider the sequence $X_1, X_2, \ldots, X_n, \ldots$ such that

$$\mathbb{P}(X_1 = 1) = 1,$$

$$\mathbb{P}(X_n = 1) = 1 - \frac{1}{n^2}, \quad \mathbb{P}(X_n = n) = \frac{1}{n^2} \quad n \geqslant 2$$

Let us prove that the sequence converges almost surely to 1.

Example

We have that

$$P_n(\epsilon) = \mathbb{P}(|X_n - 1| > \epsilon) = \begin{cases} 0, & n = 1 \\ \dfrac{1}{n^2}, & n \geqslant 2 \end{cases}$$

Therefore

$$\sum_{n=1}^{\infty} P_n(\epsilon) = \sum_{n=2}^{\infty} \frac{1}{n^2} < \infty$$

Hence

$$X_n \xrightarrow{a.s.} 1$$

Operations with limits

Theorem

- *If* $X_n \xrightarrow{a.s.} X$ *and* $Y_n \xrightarrow{a.s.} Y$, *then* $X_n + Y_n \xrightarrow{a.s.} X + Y$.
- *If* $X_n \xrightarrow{\mathbb{P}} X$ *and* $Y_n \xrightarrow{\mathbb{P}} Y$, *then* $X_n + Y_n \xrightarrow{\mathbb{P}} X + Y$.
- *If* $X_n \xrightarrow{r} X$ *and* $Y_n \xrightarrow{r} Y$ *for some* $r > 0$, *then* $X_n + Y_n \xrightarrow{r} X + Y$.

Cramer's Theorem

Theorem (Cramer)

Suppose that $X_n \xrightarrow{d} X$ *and* $Y_n \xrightarrow{\mathbb{P}} a$*, where* a *is a constant. Then*

- $X_n + Y_n \xrightarrow{d} X + a.$
- $X_n - Y_n \xrightarrow{d} X - a.$
- $X_n \cdot Y_n \xrightarrow{d} X \cdot a.$
- $\dfrac{X_n}{Y_n} \xrightarrow{d} \dfrac{X}{a}$, *for* $a \neq 0.$

Operations with limits

Theorem

Let $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$. If X_n and Y_n are independent random variables for all n and, moreover, X and Y are independent, then $X_n + Y_n \xrightarrow{d} X + Y$.

Proof: It suffices to prove that $M_{X_n+Y_n}(\omega) \to M_{X+Y}(\omega)$ as $n \to \infty$. By the convolution theorem, this reduces to proving that

$$M_{X_n}(\omega)M_{Y_n}(\omega) \to M_X(\omega)M_Y(\omega),$$

and this fact is a consequence of $M_{X_n}(\omega) \to M_X(\omega)$ and $M_{Y_n}(\omega) \to M_Y(\omega)$.

Continuous functions

Theorem

Let $X_n \xrightarrow{\mathbb{P}} a$, where a is a constant. Suppose, further, that g is a continuous function at point a. Then

$$g(X_n) \xrightarrow{\mathbb{P}} g(a) \quad \text{as } n \longrightarrow \infty.$$

Continuous functions

Proof:

Given $\epsilon > 0$ there exists $\delta > 0$ such that $|g(x) - g(a)| < \epsilon$ if $|x - a| < \delta$. Hence, for such ϵ and δ we have the inclussion:

$$\{|g(X_n) - g(a)| \geqslant \epsilon\} \subseteq \{|X_n - a| \geqslant \delta\}$$

Thus,

$$\mathbb{P}(|g(X_n) - g(a)| \geqslant \epsilon) \leqslant \mathbb{P}(|X_n - a| \geqslant \delta)$$

Therefore, since $\mathbb{P}(|X_n - a| \geqslant \delta) \to 0$, because $X_n \xrightarrow{\mathbb{P}} a$, we conclude that

$$\mathbb{P}(|g(X_n) - g(a)| \geqslant \epsilon) \to 0 \quad \text{as } n \longrightarrow \infty$$

that is to say,

$$g(X_n) \xrightarrow{\mathbb{P}} g(a) \quad \text{as } n \longrightarrow \infty$$

Delta method

Theorem

Let

- $\{a_n \colon n \geqslant 1\}$ *be a sequence of real numbers such that* $a_n \to \infty$ *as* $n \to \infty$, *and* $a_n \neq 0$ *for all* n,
- $\{X_n \colon n \geqslant 1\}$ *be a sequence of random variables an* θ *be a real number such that* $a_n(X_n - \theta) \xrightarrow{d} N(0, \sigma)$,
- g *be a real function with a continuous derivative in an interval that contains* θ *and such* $g'(\theta) \neq 0$.

Then

$$a_n \left(g(X_n) - g(\theta)\right) \xrightarrow{d} \mathrm{N}\left(0, |g'(\theta)|\, \sigma\right)$$

Shorokhod's representation theorem

Theorem

Let $X_1, X_2, \ldots, X_k, \ldots,$ and X be random variables such that $X_n \xrightarrow{d} X$ as $n \to \infty$.

Then there exists a probability space $(\Omega', \mathcal{F}', \mathbb{P}')$ and random variables $Y_1, Y_2, \ldots, Y_k, \ldots,$ and Y, which map Ω' into $\mathbb{R}$, such that:

- *$Y_1, Y_2, \ldots, Y_k, \ldots,$ and Y have the same distribution functions than $X_1, X_2, \ldots, X_k, \ldots,$ and X, respectively.*
- *$Y_n \xrightarrow{a.s.} Y$.*

Borel-Cantelli lemmas

Given a sequence of events $A_1, A_2, \ldots, A_n, \ldots$, let

$$B_n = \bigcup_{k=n}^{\infty} A_k, \quad n \geqslant 1$$

Notice that

$$B_1 \supseteq \cdots \supseteq B_n \supseteq B_{n+1} \supseteq \cdots$$

is a decreasing sequence of events. The limit of this sequence is defined as

$$\lim_{n\to\infty} B_n \equiv \bigcap_{n=1}^{\infty} B_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$$

Borel-Cantelli lemmas

Definition

Given a sequence of events $A_1, A_2, \ldots, A_n, \ldots$, the event $A^\star$ defined by

$$A^\star = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$$

is called the limit superior of the sequence and is denoted by $\limsup_n A_n$.

- Notice that $\omega \in A^\star$ if and only if ω belongs to infinitely many of the events A_n. Hence, the event $\limsup_n A_n$ happens if and only if infinitely many of the events A_n occur.

Borel-Cantelli lemmas

Theorem (Borel-Cantelli lemmas)

Let $A_1, A_2, \ldots, A_n, \ldots$ be a sequence of events and let $A^\star$ be its limit superior.

- *If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A^\star) = 0$.*
- *If the events $A_1, A_2, \ldots, A_n, \ldots$ are independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A^\star) = 1$.*

Corollary (zero-one law)

Let $A_1, A_2, \ldots, A_n, \ldots$ be a sequence of independent events and let $A^\star$ be its limit superior. Then either $\mathbb{P}(A^\star) = 0$ or $\mathbb{P}(A^\star) = 1$ according as $\sum_{n=1}^{\infty} \mathbb{P}(A_n)$ converges or diverges, respectively.

Borel-Cantelli lemmas

Let us consider the proof of the first Borel-Cantelli lemma:

$$\mathbb{P}(A^\star) = \mathbb{P}\left(\lim_{n\to\infty} B_n\right) = \lim_{n\to\infty} \mathbb{P}(B_n)$$

$$= \lim_{n\to\infty} \mathbb{P}\left(\bigcup_{k=n}^{\infty} A_k\right) \leqslant \lim_{n\to\infty} \sum_{k=n}^{\infty} \mathbb{P}(A_k) = 0,$$

because $\sum_k \mathbb{P}(A_k)$ converges by hypothesis, and so

$$\sum_{k=n}^{\infty} \mathbb{P}(A_n) \to 0 \quad \text{as } n \to \infty$$

Examples

Example 1.

Let $X_1, X_2, \ldots, X_n, \ldots$ be a sequence of random variables such that

$$\mathbb{P}(X_n = 0) = \frac{1}{n^2}, \quad \mathbb{P}(X_n = 1) = 1 - \frac{1}{n^2} \quad n \geqslant 1$$

Let $A_n = \{X_n = 0\}$, $n \geqslant 1$.

- Since $\sum_n \mathbb{P}(A_n) = \sum_{n=1}^{\infty} 1/n^2 < \infty$, the first Borel-Cantelli lemma implies that $\mathbb{P}(A^\star) = 0$. Thus, the probability that $\{X_n = 0\}$ happens infinitely often is equal to 0.
- We conclude that $\mathbb{P}(X_n = 1 \text{ for all } n \text{ suficiently large}) = 1$, and hence

$$X_n \xrightarrow{a.s.} 1$$

Examples

Example 2.

Now, let $X_1, X_2, \ldots, X_n, \ldots$ be independent random variables such that

$$\mathbb{P}(X_n = 0) = \frac{1}{n}, \quad \mathbb{P}(X_n = 1) = 1 - \frac{1}{n} \quad n \geqslant 1$$

- Let $A_n = \{X_n = 0\}$. Since the events A_n are independent, and $\sum_n \mathbb{P}(A_n) = \sum_{n=1}^{\infty} 1/n = \infty$, the second Borel-Cantelli lemma indicates that $\mathbb{P}(X_n = 0 \text{ occurring infinitely often}) = 1$.
- Analogously, let $\overline{A}_n = \{X_n = 1\}$. The series $\sum_n \mathbb{P}(\overline{A}_n)$ also diverges. Thus, $\mathbb{P}(X_n = 1 \text{ occurring infinitely often}) = 1$.
- We conclude that, almost surely, the sequence $\{X_n \colon n \geqslant 1\}$ does not converge.

Examples

Example 3.

Let a fair coin be tossed infinitely many times and denote by $A^\star$ the event "heads occurs infinitely often".

Thus, $A^\star = \limsup_n A_n$, where A_n is the event "the n-th toss lands heads".

- Since the events A_n are independent, the second Borel-Cantelly lemma implies $\mathbb{P}(A^\star) = 1$, because $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \sum_{n=1}^{\infty} 1/2 = \infty$.
- What happens if $\mathbb{P}(\text{heads}) = \epsilon$, being $\epsilon > 0$ a very small number. Is it still true that heads will occur infinitely often?

  Again, $\mathbb{P}(A^\star) = 1$, because $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \sum_{n=1}^{\infty} \epsilon = \infty$.

Examples

Example 4.

Now, let $\mathcal{P}_k$ be a given pattern of heads and tails of length k. For instance, $\mathcal{P}_6 = \text{HHTTHT}$.

Fact

If a coin is tossed infinitely many times, the probability of observing $\mathcal{P}_k$ infinitely often is 1, no matters how large is k.

Proof:

Let A_n be the event "the pattern $\mathcal{P}_k$ is observed between the $[(n-1)k+1]$-th and the $[nk]$-th toss, $n \geqslant 1$".

- If A_n happens infinitely often, then it is true that the pattern $\mathcal{P}_k$ is observed infinitely often. So to prove tha fact it suffices to prove that $\mathbb{P}(\limsup_n A_n) = 1$

Examples

- The events A_n are independent, because if $i \neq j$, then A_i and A_j correspond to two non-ovelapping sequences of tosses.
- If the pattern $\mathcal{P}_k$ contains r heads and $\mathbb{P}(\text{heads}) = p$, $0 < p < 1$, then $\mathbb{P}(A_n) = \epsilon$, where $\epsilon \equiv p^r(1-p)^{k-r} > 0$.
- Observe that if k is large, ϵ is very small but nonzero. Again, $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \sum_{n=1}^{\infty} \epsilon = \infty$, and the second Borel-Cantelli lemma implies
$$\mathbb{P}(\limsup_n A_n) = 1$$
- So, $\mathcal{P}_k$ will be almost surely observed infinitely many times.

The infinite monkey theorem

Theorem

A monkey hitting keys at random on a typewriter keyboard for an infinite amount of time will almost surely type any given text, such as the complete works of William Shakespeare.

In fact, the monkey would almost surely type every possible finite text an infinite number of times.

Proof: see Wiquipedia:

`https://en.wikipedia.org/wiki/Infinite_monkey_theorem`